

ZERO DCC

DIGITAL COMMAND AND CONTROL INTERFACE FOR THE HORNBY ZERO 1 TRAIN CONTROLLER

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The Zero DCC module provides a significant upgrade to Zero 1 users. To use this interface effectively, you must have a Zero 1 Master Controller and DCC equipped and programmed locomotives and points/turnouts.

The interface not only provides an upgrade allowing the use of your existing Zero 1 functions to command a modern DCC system, but it also provides additional DCC functions, not previously available using Zero 1.

Warning: Zero 1 equipped locomotives and accessory modules will likely be damaged using this equipment. The only exception is ZTC decoders that are both Zero 1 and DCC compatible.

**DO NOT USE ZERO 1 EQUIPPED LOCOMOTIVES ON TRACKS CONNECTED
TO THIS INTERFACE. YOU HAVE BEEN WARNED!!!!**

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Commands

All the usual Zero 1 commands should work as before.

- Locomotive select
- Inertia select
- Slave select
- Throttle control
- Forward/Reverse
- Turnouts 1-89 direction (NB: 90-99 are used for new features)
- Panic (NB: Panic only puts all locomotives to zero speed – the track is still live!)

The Zero DCC interface provides the following additional features:

- Control track power, on/off etc. (Accessory 90)
- Program DCC decoder ID (CV1) on a separate program track (Accessory 91)
- Record and replay a Zero 1 session (Accessory 95)
- Selection of up to 16 DCC locomotives for control – can be any 16 from your DCC roster! (Accessory 96)
- Save and recall different rosters of 16 locomotives on built in SD card (Accessory 97)
- Simple head light and whistle/horn control (Accessory 98)
- Separate full control of the DCC 29 CAB functions (Accessory 99)
- USB connection for computer access via JMRI and other software
- WiFi connection for computer or mobile access
- Large LCD screen to display your locomotive roster, speed and direction

Wiring

The Zero DCC interface box has several power inlets and outlets.

Interface power

You will require the usual power requirements of the Zero 1 Phase 1 & 2 Master Controller and Phase 3 Micromimic Console.

In addition to the Zero 1 power, the interface requires two power sources and access to the Zero 1 track power (inputs):

Inputs	Comment
USB connector	This is for USB connection of the DCC-EX module to a computer for JMRI control and programming
12V DC via 5.2 plug	This is to provide module and track power (can be up to 5A)
Zero 1 track power (22V AC)	This is the Zero 1 input signal
Outputs	Comment
12V DCC to Track A (main track by default)	This provides the DCC signal to the main track to control your trains.
12V DCC to Track B (programming track)	This provides low current DCC to a special programming track to allow you to set your locomotive number or program via JMRI
Switched Zero 1 track power (22A AC)	This is provided for those using the Zero 1 Phase 3 Micromimic system. This is a direct feed from the Zero 1 Master Controller but is switched so that the additional functions of the Zero DCC interface don't interfere with the Micromimic's panel display output.

Track Power (Accessory 90)

While the Zero 1 green power light will come on when the Zero 1 has mains power available, that does not mean that the track will be getting power from the Zero DCC interface.

The track settings and status can be seen on the top of the Zero DCC LCD screen.

Function 90 provides access to the track power functions.

To enter function 90 select '9 - 0 - →'

To exist function 90, select '9 - 0 - ←'

By default the track power functions are set to:

Track A:	Main	(ie main track running)
Track B:	Program	(ie second track for DCC programming)
Power:	Off	

WARNING: Before turning track power on, please ensure that any locomotives and other powered items are suitable for the settings selected (DCC by default).

Options under function 90:

- Option 1: Turn track power on/off: select "1- →"
- Option 2: Track A = DCC Main; Track B = DCC Program (default)
- Option 3: Track A = DCC Main; Track B = 12V DC CAB No. 999
- Option 4: Track A = 12VDC CAB No. 999; Track B = 12V DC CAB No. 998

NB: Whenever you change the track settings (ie Option 2 – 4), the track power will be turned off to protect any locomotives you have on the track that do not match the setting chosen.

Panic!

Zero 1's old Panic function (the red button) will still operate as before.

When pushed, the panic function instantly turns the speed of all locomotives to zero.

To cancel a Panic situation, press the 'C' key.

It does not cut the power to the track. Accessory 90 will do that (see above).

Errors

If you happen to mis-key any default Zero 1 commands, the yellow Error light will illuminate.

To clear the Error state, press the 'C' key.

Unfortunately, if you mis-key any of the Zero DCC interface commands, they will just not work, or give you un-expected results.

Remember, that Zero DCC interface Accessory functions 99 and 98 can be turned off by typing '99 - ← - ←' and '98 - ← - ←' respectively.

Overload

With the use of the Zero DCC interface, hopefully you will never see the Overload red light illuminate. If you do, it is likely because of a short circuit in the Zero 1 track or accessory power outputs.

Ensure that there are no short circuits and press the 'C' key to clear the message.

If it continues, turn off your Zero 1 and the Zero DCC interface and check all the wiring again.

If the Zero DCC module overloads, track power will be turned off (check LCD display) and power will be cut to all tracks connected to the DCC outputs. If this does not resolve, check for shorts on the track and in your layout wiring.

Locomotive control

To select a locomotive from the 16 on your list, select 'Loco – *number_to_16* - →'

(in this case you can use either arrow to finish the sequence).

Once you have selected the loco, then the Master Controller slide controller (Speed Regulator) will provide throttle and the FORWARD and REVERSE buttons will control the direction of travel.

Slave controllers

You can have up to three slave controllers by default. This includes one hand held slave controller. Controllers are numbered from the left to right (ie the Master Controller is Slave number 1).

For the more advanced user, with a little bit of modification, you can have up to 15 slave controllers. With the Master Controller that means you can have one controller per locomotive!

To use a slave controller, select:

'Slave – *number_of_slave* – Loco – *number_of_loco* - →'

The selected slave will then have control of the locomotive nominated.

Inertia

Zero 1's in built inertia control will continue to function if desired. NB: This is in addition to any inertia you have pre-programmed into your locomotive's DCC decoder.

Zero 1 has four inertia levels from minimum 1 to maximum 4.

To use Zero 1's inertia, select 'INERTIA – *number_from_1_to_4* – Slave – *number_of_slave* Loco – *number_of_loco* - →'

The selected slave will then have control of the locomotive nominated with the level of inertia selected.

Point/Turnout control

Due to the new features added to the Zero 1 interface to DCC, the number of points able to be controlled has been reduced to 89. Accessories 90-99 are used for other functions.

If access to a larger number of points or accessories is needed, please contact the provider of the interface for a special build.

To change the state of DCC equipped points 1-89, select:
'point_number - →' (or *'←'*)

Micromimic Console and Displays

If you have a Zero 1 Phase 3 Micromimic Console and Display, it should function as normal.

However, to overcome some issues that the Zero DCC interface might cause, rather than connecting it directly to the Zero 1 Master Controller, the Zero DCC interface is supplied with a special output for the Micromimic Console.

This output is switched off by the Zero DCC interface when ever a special accessory function (ie 90-99) is used so that the Micromimic Display continues to show the correct alignment of points while these functions are used. The Zero 1 output is switched back to the Micromimic Console when these functions have ended.

An option is to include Micromimic LEDs for accessories 90-99 placed to indicate the status of these functions on the Micromimic Display Panels (eg to show that Accessory 90 for track power is on or off).

To record a Zero 1 session for play back (Accessory 95)

You can record the movement and function control of locomotives and points to SD card and play them back at any time.

It is important to remember, that the timing will be based on the conditions of your track at the time of the recording. Any change to the maintenance of locomotives, points or the state of the track (including cleaning) will likely effect the timing sequence of the recording or playback. So make sure your layout is operating correctly before making any recording, or playback.

Please also remember that once a recording has commenced re-playing, you will not have access to the Zero 1 Panic button or the Accessory 90 track power function! This means if anything goes wrong, you will need to turn off the power to the layout at the wall switch!

You will also loose access to some other Zero 1 or Zero DCC functions during a recording play back.

Recording

1. Before you start, make sure your locomotives, track and points are all operating efficiently
2. To select the recording function, turn the accessory on, select '9 - 5 - →'
3. Select the number to set the recording to (1-89) and press the right arrow,
eg '1 - →'

From now on, any changes made to locomotive control, locomotive functions and point position will be recorded to SD card until you turn the recording off.

4. To end the recording, select '9 - 5 - →' again

Example: You want to make a recording of the locomotive leaving a siding, make a loop of the track and stop at a station:

Make sure track and locomotive are in good condition before you start

9 - 5 - → *enter recording mode*

5 - → *select to make recording number 5*

Use the Zero 1 to open the siding point, take the train out, close the point again, travel around the loop and stop at the station.

9 - 5 - → *to turn off the recording*

Playback

1. To play back, while still in the recording function (95), select the number of the recording to play back (1-89) and press the left arrow, eg select '1 - ←'.

While the play back is in progress, you can still control your locomotives and points, but remember the recording will not adjust for any changes you make, it will continue as originally recorded!

2. If you need to end the play back early, then select '9 - 5 - →'. You will still be in function 95 mode.
3. To exit function 95, select '9 - 5 - ← - ←' (yes, press the left arrow twice!)

Now return the train to the starting position in the siding.

5 - ←

to replay the recording

9 - 5 - ← - ←

to exit recording mode

To change the locomotives in your control list (Accessory 96)

Zero 1 only allows control of 16 locomotives at a time. By default they are set to DCC decoder 1-16. However, you can change them at any time to any locomotive in your roster. Once changed, they are automatically stored on a backup SD card and restored on your next session.

Zero 1 does not allow the use of zero (0) as an accessory. If a zero is needed in your DCC command sequence, use 10 as a work around (see example below.)

1. To turn the list control accessory on, select '9 - 6 - →'

To turn the list control accessory off, select '9 - 6 - ← - ←' (yes, press the left arrow twice!)

2. To change a locomotive in your list:
 - a. Enter the DCC CAB ID one digit at a time with a '←' to confirm each digit
NB: to enter a 0, use 10 instead!
 - b. To add this locomotive to the control list, enter the desired number from the control list (1-16) and confirm with →'
3. You can change as many locomotives in the control list as you like while the accessory is turned on!
4. To turn the accessory off, select '9 - 6 - ← - ←'

Example: If you wanted to add a locomotive with the DCC CAD ID of 909 to Zero 1 control list number 14 (ie locomotive number 14) the sequence would be:

9 - 6 - →

to turn on the accessory

9 - ← - 10 - ← - 9 - ←

to enter the DCC CAD ID (remember 0's are 10)

14 - →

to enter the Zero 1 control list number to assign

you can change as many as you like, then

9 - 6 - ← - ←

turn off the accessory

To set up a new list of locomotives, or recall a previous one (Accessory 97)

Accessory 96 (above) allows you to set which DCC locomotive ID is set up in Zero 1's list of controllable locomotives, and saves it on SD card.

Accessory 97 allows you to set up multiple lists of 16 locomotives.

So you could have a list of:

- 16 favourite locomotives
- 16 least favourite locomotives
- 16 steam locomotives
- 16 diesel locomotives
- 16 locomotives from the 1920's
- 16 locomotives from the 2020's
- And so on....

To use this function, remember that when changing a locomotive in Accessory 96 (above), that the change is automatically changed in the current list.

If you haven't created a new list, Accessory 97 then takes the current list and assigns it a new number (1-10), then any changes made in Accessory 96 are recorded in the new list.

If you have already created a new list, Accessory 97 will load it from the SD card for you.

1. To turn the list control accessory on, select '9 - 7 - →'
2. To change a locomotive in your list, then enter the list number you wish to create or recall (1-10) and confirm with →'
3. To turn the accessory off, select '9 - 7 - ← - ←'

Example: If you wanted to add a new list numbered 3:

9 - 7 - →

to turn on the accessory

3 - →

to enter the list number to assign

9 - 7 - ← - ←

turn off the accessory (technically not needed as this accessory will turn itself off after a list is chosen)

Remember, if you already have a list '3' created this will load it for you. If you don't have a list '3' this will create it based on the current locomotive roster in place and will be saved after you make the first change using Accessory 96 (above).

Head light and whistle/horn control (Accessory 98)

NB: You must have a sound equipped and programmed DCC locomotive for the whistle/horn to work, and a headlight equipped and programmed DCC locomotive for the headlight to work!

1. To turn the head light and whistle/horn accessory on, select '9 - 8 - →'

To turn the head light and whistle/horn accessory off, select '9 - 8 - ← - ←'

2. Once the function is turned on, select the number of the locomotive
3. To turn the head light on and off, select '→'
4. To turn the whistle/horn on and off, select '←' (NB: remember it will keep going until you toggle it back off!)

While the locomotive is selected you can keep toggling the head light and whistle/horn on/off as many times as you like.

While the accessory is on, you can change the number of the locomotive at any time.

To turn the function off, select '9 - 8 - ← - ←'

Example: If you wanted to blow the whistle on a sound and light equipped DCC locomotive number 12 and turn on the headlight the sequence would be:

9 - 8 - →
12
→
→
←

*to turn on the function
to enter the locomotive number
to blow the whistle
to make it stop!
to turn on the headlight*

you can blow the whistle, and turn on and off the headlight as many times as you like, until you change the locomotive number or turn off the accessory

you can change loco numbers as many time as you like, until you turn off the accessory

9 - 8 - ← - ←

turn off the accessory

DCC CAB function controls (Accessory 99)

NB: You must have a decoder programmed for multiple functions for this to work!

1. To turn the CAB functions accessory on, select '9 - 9 - →'

To turn the CAB functions accessory off, select '9 - 9 - ← - ←'

2. Once the accessory is turned on, select the number of the locomotive
3. Next confirm with '→'
4. Next select the number of the CAB function (eg whistle = 2)
5. To turn the function on and off, select '→'

The locomotive will remain selected (until you turn Accessory 98 off). While the current function is selected you can keep toggling it on/off as many times as you like.

While the accessory is on, you can change the function number at any time.

To turn the function off, select '9 - 9 - ← - ←'

Example: If you wanted to blow the whistle on a sound and light equipped DCC locomotive number 12 the sequence would be:

9 - 9 - →

12 - →

2 - →

→

to turn on the function

to enter the locomotive number

to toggle function 2 (ie blow the whistle)

to toggle it again

you can change between functions on the selected locomotive as many time as you like, until you turn off the accessory

9 - 9 - ← - ←

turn off the accessory

TIP: If your DCC decoder requires you to 'start up' the locomotive with CAB Function 8, use this Zero DCC accessory function (just replace the '2' in the above example with an '8'!)

JMRI: A Java Model Railroad Interface

The Zero DCC interface also provides a USB port to connect to your computer to provide an interface with the open source JMRI applications and associated functions.

But it's much more fun controlling your trains with the Zero 1!

Further JMRI information and downloads are available at: <https://www.jmri.org/>

WiFi access for computer or mobile device

The Zero DCC interface is also capable of WiFi Connections for JMRI WiThrottle compatible applications (eg Android Engine Driver or iOS WiThrottle).

The Zero DCC interface will operate as an independent WiFi access point. On your device, connect to the access point with the name commencing with "DCC" the remainder of the name is linked to your individual interface's network MAC address.

The password for your device is "Password" plus the MAC address that was part of your interface's identifier.

Once you are connected to your interface's local WiFi network, you can use your JMRI WiThrottle application to control your locomotives and points.

In your mobile application, you will need to point it to the following:

IP address: 192.168.0.4
Port: 2560

Remember, you are accessing the DCC controller directly, and can only control your DCC locomotives and any DCC wired points – you cannot change your Zero 1 points or accessories using the WiThrottle; and your Micromimic Display will not display any point changes.

To program the DCC decoder ID (CV1) on a separate program track (Accessory 91)

Warning: This feature of the Zero DCC interface comes with some risk!

Power interruptions, timing issues and other unforeseen problems may damage your decoder. Consider having your locomotive programmed by a professional.

USE THIS FEATURE AT YOUR OWN RISK – YOU HAVE BEEN WARNED!!!!

If you choose to proceed, please be very careful and double check everything you are planning to do.

You cannot do this on the main track. You will need a separate programming track connected to the programming track outputs on the Zero DCC interface.

Consider the DCC locomotive ID you wish to use. Some things to consider:

- All decoders come with a default DCC ID of 3. So best to avoid that.
- Using a DCC ID of zero (0) can cause problems with your configuration – so the Zero DCC interface has been written to prevent you doing this.
- There are a number of DCC systems that will not recognise a DCC ID over 99 – so the Zero DCC interface has been written to prevent you doing this (so that if you take your locomotives to another layout using one of these systems it will still work).

Please read the following instructions several times and work through what you will be doing BEFORE attempting this RISKY procedure.

Remember on the Zero 1 you will not have any acknowledgement that the key press was recognised.

You will need to be planned and organised; slow and deliberate.

Write out your key press sequence in advance to help prevent mistakes.

READ THIS SEVERAL TIMES:

1. Prepare for the process and ensure that power, track connections or any other issues that will interfere with the process are prevented.
 2. Set up your programming track.
 - It cannot be connected to the main line in any way.
 - The locomotive may move during programming, so ensure the track is in a safe location. Consider buffers at each end.
 - Connect the programming track to the programming track outlets on the Zero DCC interface.
 3. DO NOT PUT YOUR LOCOMOTIVE ON THE PROGRAMING TRACK YET
 4. Turn the power to the track OFF, select '9 - 0 - ←'
 5. Confirm that the track power is off
 6. Now, place the locomotive you wish to program onto the separate programming track.
 7. Turn the power to the track on, select '9 - 0 - →'
 8. Turn on the programming accessory on, select '9 - 1 - →'
 9. Now carefully enter the new DCC ID you want to program:
 - a. Enter the DCC CAB ID one digit at a time
 - b. Confirm each digit with a '←'.
- NB: to enter a zero (0), use 1-0 instead!
10. Up to this point, if you make any mistakes you can cancel the entire programming routine by selecting '9 - 1 - ←'
 11. When you have finished carefully entering the new DCC ID, to commence the programming sequence, press the '→' key
 12. Remember, the locomotive may move while being programmed so be prepared. DO NOT ALLOW IT TO LEAVE THE TRACK!
 13. As the Zero 1 has no feedback mechanism to advise you that this was completed, wait a few minutes before removing the locomotive. You may hear some relays click in the Zero DCC interface when the process has completed, but still wait a few minutes before removing the locomotive.
 14. You do not need to end the programming routine, it will end automatically after the programming commands have been sent to the DCC base station (or if you key any incorrect sequences. Eg using a DCC CAB ID of zero.) But you

can still turn off the programming routine by selecting '9 - 1 - ←' to confirm. If you are using a Micromimic Display and have an LED set up for accessory 96 you will see it change – that is an indication that the DCC programming is complete.

15. Test your locomotive on the main line. It should now respond to the new DCC CAB ID – remember to add it to your roster (Accessory 96) before you try testing it.

16. Read all of this again, and write down your key sequence.

17. Read it again.

Problem solving the programming function.

If your locomotive does not respond to its new DCC CAB ID, here are some things to consider. You will see a theme here. This process is risky using the Zero 1. Consider having your locomotive programmed by a professional.

- Did you try selecting a DCC CAB ID of zero (0) or over 99? The software deliberately prevented you from programming these. Read the manual again. Consider having your locomotive programmed by a professional.
- Does the locomotive respond to its old DCC CAB ID (most are default 3 when new)? The process didn't work. Read the manual again. Consider having your locomotive programmed by a professional.
- Could you have mis-keyed during the process? Read the manual again. Consider having your locomotive programmed by a professional.
- Did the power fail during the process? This could have damaged the decoder. Consider having your locomotive programmed by a professional.
- Did the locomotive lose contact with the programming track at any point during the process? Read the manual again. Consider having your locomotive programmed by a professional.
- Did any of the wiring connections come loose at any point during the process? Read the manual again. Consider having your locomotive programmed by a professional.
- Did you try programming the locomotive on the main track? Read the manual again. Consider having your locomotive programmed by a professional.
- Consider having your locomotive programmed by a professional.

Acknowledgements

We would like to thank Hornby Hobbies and their forebears for creating the Zero 1 control system. It was ahead of its time, and hopefully this project will demonstrate that the system is still capable in the 21st century.

We would also like to thank all the developers of the DCC++EX (<https://dcc-ex.com>) open source base station software and hardware schemes, on which this interface is based, for making such a flexible and adaptable platform for digital model railroading.

We make no claims for the excellent work of all these people over such a long time. We are indebted to you all.

We do, however, take great credit ;) for creating the interface that allows these two great systems to talk to each other and hopefully keep the Zero 1 community alive and creative in the 21st century.